Project Report

On

Digital Counter Using 555 Timer and CD4026 with 7-Segment Display

Course No: ECE 2104

Course Name: Digital Electronics and Logic Circuits

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17 May, 2026

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1. Detailed Specification of the Project

1.1 Abstract

This project presents the design and implementation of a two-digit digital counter capable of counting from 00 to 99 using a 555 timer IC in astable mode for clock pulse generation and two CD4026 decade counter ICs to drive seven-segment displays. The first CD4026 counts the units digit (0–9), and its carry output triggers the second CD4026 to increment the tens digit, enabling continuous sequential counting from 00 to 99.

Additional features were added to improve usability and control, including a pause switch to stop the counting process temporarily and a reset switch to return the counter to 00 instantly. A pulse indicator LED was connected to the timer output to visually represent each clock pulse. Another LED indicator circuit using a BC547 transistor was designed to glow at every 10 counts (10, 20, 30, etc.). A 100 kΩ potentiometer was also used to control the clock frequency and adjust the counting speed.

Overall, the project demonstrates the practical application of digital logic circuits, including clock generation, cascading counters, frequency division, and seven-segment display interfacing without using a microcontroller. The circuit was tested and analyzed to ensure proper functionality, accuracy, and stable operation.

1.2 Objectives

- To design a digital counter capable of counting from 00 to 99.
- To generate clock pulses using a 555 timer IC.
- To implement decade counting using CD4026 ICs.
- To interface seven-segment displays with digital counter circuits.
- To understand cascading techniques used in multi-digit counters.
- To implement pause and reset control functionality.
- To implement adjustable counting speed using a potentiometer.
- To design LED indicators for pulse monitoring and decade counting.
- To analyze the behavior of sequential digital circuits.

1.3 Expected Outcome

The expected output of the project is:

- Sequential counting from 00 to 99.
- Proper synchronization between the units and tens digits.
- Stable clock pulse generation using the 555 timer.
- Adjustable counting speed using the potentiometer.
- Correct display of decimal numbers on two seven-segment displays.
- LED blinking for every generated pulse.
- Additional LED glowing after every 10 pulses.
- Successful pause and reset operations.

1.4 Required Components

Serial No.	Component Name	Quantity
1	NE555 Timer IC	1
2	CD4026 Decade Counter IC	2
3	Seven Segment Display (Common Cathode)	2
4	BC547 Transistor	1
5	100 kΩ Potentiometer	1
6	Resistor (10 kΩ, 1 kΩ, 330 Ω, 220 Ω)	8
7	Capacitor (47 µF, 10 nF, 10 µF)	3
8	LED	2
9	Push Button	2
10	5V DC Supply	1
11	Connecting Wires and Breadboard	As Required

2. Methodology

2.1 Working Principle

The project consists of five major sections:

1. Clock pulse generation using the 555 timer IC.
2. Counting section using CD4026 ICs.
3. Display section using seven-segment displays.
4. User control section using pause and reset switches.
5. LED indication and transistor switching section.

The 555 timer operates in astable mode to generate continuous square wave clock pulses, which drive the first CD4026 decade counter IC. This IC counts the units digit (0–9) and directly controls the first seven-segment display. After each full cycle, its carry-out signal triggers the second CD4026, which counts the tens digit, allowing the system to count sequentially from 00 to 99.

A pause switch is used to temporarily stop the clock, while a reset button returns both counters to 00. An LED connected to the 555 timer output blinks with each pulse, and another LED indicator using a BC547 transistor lights up every 10 counts. The counting speed is adjustable using a 100 kΩ potentiometer in the timer circuit.

2.2 Operation of the 555 Timer

The 555 timer is configured in astable mode where it continuously switches between HIGH and LOW states to generate periodic clock pulses. The output frequency depends on the resistor, potentiometer, and capacitor values connected in the circuit.

The approximate frequency equation is:

$$f = 1.44/(R1+2R2)C$$

Where:

- R1 and R2 are timing resistors.
- C is the timing capacitor.
- f is the output frequency.

The 100 kΩ potentiometer changes the resistance value, thereby controlling the counting speed. These clock pulses are fed into the clock input pin of the first CD4026 counter.

2.3 Operation of the CD4026 Counter IC

The CD4026 is a decade counter with an integrated seven-segment decoder driver. It counts clock pulses and directly drives a seven-segment display without requiring additional decoder ICs. Main functions of the CD4026:

- Counts from 0 to 9.
- Converts binary count into seven-segment display outputs.
- Provides carry-out signal for cascading counters.
- Supports reset functionality.
- Drives common cathode seven-segment displays directly.

The carry-out pin of the first CD4026 is connected to the clock input of the second CD4026 to achieve two-digit counting.

2.4 Operation of Pause and Reset Controls

The pause switch temporarily disables clock pulse generation from the 555 timer, stopping the counting process while preserving the current display value.

The reset push button is connected to the reset pins of the CD4026 ICs. Pressing the button resets both counters and the display returns to 00.

2.5 Operation of LED Indicator Circuits

Pulse Indicator LED: An LED connected to the output of the 555 timer glows with every clock pulse, visually confirming pulse generation.

Decade Indicator LED: A BC547 transistor-based switching circuit was added to activate another LED after every 10 pulses. The carry-out signal from the first CD4026 triggers this transistor circuit, causing the LED to glow at counts such as: 10 → 20 → 30 → 40 → ...

3. Design of the Project

3.1 Circuit Description

The circuit uses a 555 timer IC in astable mode with resistors, capacitors, and a 100 kΩ potentiometer to generate adjustable clock pulses. The output from pin 3 drives the first CD4026 IC, which controls the units digit seven-segment display. Its carry-out is connected to a second CD4026 IC for the tens digit display, enabling counting from 00 to 99.

A pulse LED indicates each clock pulse, while a BC547-based LED circuit glows every 10 counts using the carry signal. A pause switch stops counting temporarily, and a reset button returns the display to 00.

3.2 Circuit Schematic

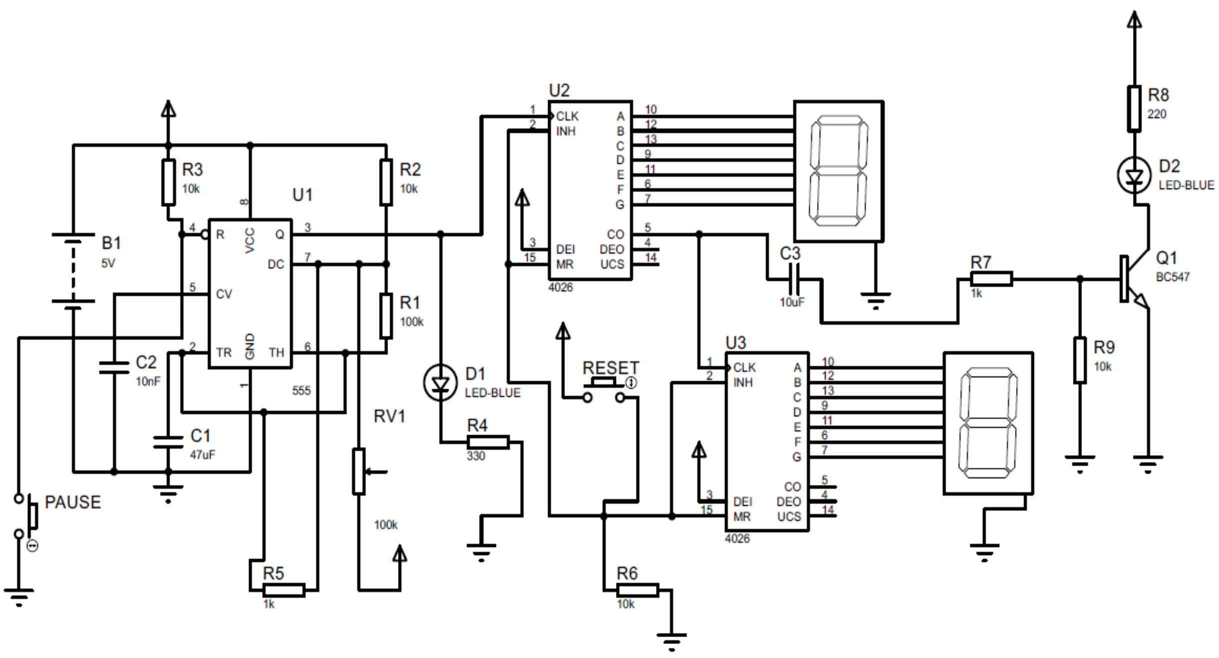

Fig: Circuit Diagram of a Digital Counter Using 555 Timer and CD4026 with 7 Segment Display

3.3 Pin Configuration and Connections

555 Timer IC

Pin Number	Function
1	Ground
2	Trigger
3	Output
4	Reset
5	Control Voltage
6	Threshold
7	Discharge
8	VCC

CD4026 IC

Pin Number	Function
1	Clock Input
2	Clock Inhibit
3	Display Enable
5	Carry Out
15	Reset
10–13, 9, 11, 6, 7	Seven Segment Outputs

3.4 Timing Diagram

The timing sequence of the counter operation is based on periodic clock pulses generated by the 555 timer.

- Each pulse increments the first counter.
- After every 10 pulses, the second counter increments by one.
- The decade indicator LED glows after every 10 pulses.
- The sequence continues until the display reaches 99.

Example counting sequence:

00 → 01 → 02 → ... → 09 → 10 → 11 → ... → 99

3.5 Flow of Operation

Power supply is applied to the circuit
↓
555 timer generates adjustable clock pulses
↓
Pulse LED blinks for each pulse
↓
CD4026 (1st counter) counts incoming pulses
↓
First seven-segment display updates continuously
↓
After every 10 counts, carry output triggers second CD4026
↓
Second seven-segment display increments by one digit
↓
Decade indicator LED glows after every 10 pulses
↓
Pause switch can stop counting temporarily
↓
Reset button resets display to 00
↓
Process repeats continuously

4. Result Analysis

The circuit successfully counted from 00 to 99 using two seven-segment displays. The 555 timer produced stable, adjustable clock pulses, confirmed by the pulse indicator LED blinking according to the frequency. The first CD4026 correctly counted the units digit from 0 to 9, and its carry-out signal successfully triggered the second CD4026 for the tens digit. The decade indicator LED glowed at every 10 counts (10, 20, 30, etc.), verifying proper carry-out operation and transistor switching. The pause switch stopped the counting process as intended, and the reset switch instantly returned the display to 00.

The following observations were made during simulation and implementation:

Observation	Result
Clock pulse generation	Successful
Adjustable speed control	Successful
Units digit counting	Accurate

Observation	Result
Tens digit counting	Accurate
Display operation	Stable
Counter cascading	Successful
Pause functionality	Working properly
Reset functionality	Working properly
Pulse LED indication	Successful
Decade LED indication	Successful
Power supply operation	Stable at 5V

5. Problems and Technical Issues Encountered

Several practical and technical issues were encountered during the development of the project.

5.1 Unstable Clock Pulses

Improper resistor, potentiometer, and capacitor values caused unstable counting speed. Adjusting the timing components stabilized the clock signal.

5.2 Potentiometer Speed Control Issue

Initially, the counter speed remained excessively fast even after rotating the potentiometer. This happened because the resistance range and timing capacitor values were not properly balanced. The issue was reduced by adjusting resistor-capacitor combinations and using suitable timing capacitor values.

5.3 Wiring Complexity

The large number of connections between ICs, displays, LEDs, switches, and transistor circuits increased wiring complexity. Proper labeling and organized wiring improved circuit cleanliness.

5.4 Power Supply Noise

Minor display flickering was observed due to unstable supply conditions. Using a regulated 5V supply improved the performance.

5.5 Pause Button Issue: When the pause button was pressed for too long, the counter sometimes skipped counts and jumped to higher digits. This happened due to switch bouncing and unstable triggering in the button circuit.

6. Conclusion

The project successfully implemented a two-digit digital counter (00–99) using a 555 timer IC for adjustable clock pulse generation and CD4026 decade counter ICs for accurate counting and seven-segment display control. Additional features such as pause and reset functions, pulse and decade indicator LEDs, transistor switching, and potentiometer-based speed control enhanced the system's usability and performance.

The project demonstrated important digital electronics concepts, including:
• Sequential logic circuits
• Pulse generation using 555 timer IC
• Decade counting
• Cascading of counter ICs
• Seven-segment display interfacing
• Transistor switching operation
• Frequency control using potentiometer

The final circuit operated according to the required specifications and successfully achieved the intended objective.

7. Limitations

- The counting speed still depends on RC timing components.
- The circuit only counts up to 99.
- High counting speed may make display changes difficult to observe.
- Breadboard wiring increases chances of loose connections.
- The brightness of LEDs and displays may vary with supply conditions.

8. Future Improvements

The project can be improved further by:

- Adding buzzer indication at specific counts.
- Extending the counter beyond 99 using additional CD4026 ICs.
- Using a crystal oscillator for more accurate timing.
- Designing the circuit on a PCB for better reliability.
- Adding up/down counting functionality.
- Adding memory storage for count values.
- Implementing automatic stop at a selected count.
- Using a microcontroller for advanced control and display options.

9. Applications

This type of digital counter can be used in:

- Event counting systems
- Digital scoreboards
- Object counting machines
- Production line monitoring
- Educational demonstration circuits
- Industrial automation systems
- Frequency counting applications
- Traffic counting systems

10. GitHub Repository

GitHub Repository Link:

References

1. NE555 Timer IC Datasheet.
2. CD4026 Decade Counter IC Datasheet.
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5. Floyd, T. L., *Digital Fundamentals*.
6. Malvino, A. P., *Digital Computer Electronics*.